

CONTACT

-
- linkedin.com/in/paddyiyer
- paddyspeaks.com
- github.com/paddyriyer
- San Francisco Bay Area

TECHNICAL SKILLS

Spark

Hive

Hadoop

Databricks

Delta Lake

Snowflake

Redshift

dbt

Airflow

Datascwarm

AWS

Azure

GCP

Terraform

Kubernetes

Python

SQL

Scala

LLM/RAG Pipelines

Claude Code

AI Agents

Prompt Eng.

LangChain

GitHub Copilot

MLflow

Diff. Privacy

Unity Catalog

Data Mesh

EDUCATION

- Data Engineering Cert.**
UC Santa Cruz, 2013–14
- B.S. Electrical Engineering**
Sardar Patel College of Eng.
- Diploma, Electronics & Comms**
Technical Board, Tamil Nadu

CERTIFICATIONS

- VMware SaaS Essentials
- Hadoop Fundamentals
- NoSQL Databases
- NoSQL for SQL Professionals

PUBLICATIONS

- Cloud Computing
- All About Big Data
- Future Trends in BI

AI & THOUGHT LEADERSHIP

- 128 articles on paddyspeaks.com
- Claude Code skill development
- Vibe coding methodology
- AI agent architecture writing

OPEN-SOURCE & COMMUNITY

- Interview Studio — free Data & AI Engineering prep
- 22 data-model deep-dives + 1,500+ questions
- In-browser SQL / Python (WASM), no backend

Paddy Iyer

DATA ENGINEERING LEADER · STRATEGIC DATA ARCHITECT · AI & PRIVACY ENGINEERING · AI INNOVATOR

35+ years building and transforming enterprise data platforms for global tech leaders across cloud/SaaS, fintech, gaming, and consumer platforms. Expert in cloud-native architectures, real-time analytics, privacy-first pipelines, and AI-augmented data operations. Proven builder of high-trust, high-performance data teams at petabyte scale. Published writer and thought leader on data architecture, AI agents, and the intersection of ancient wisdom with modern technology.

Modern Data Architecture

Privacy-First Design

Consent Architecture

AI Agents & LLM Pipelines

Prompt Engineering

Data Governance (C1–C5)

Petabyte-Scale Systems

Global Team Leadership

Data Storytelling

FEATURED EXPERIENCE

Data Engineer — Privacy & Consent Infrastructure

Apr 2024 – Present

Meta

- Delivered ~\$500K/year in recovered engineering capacity by sustaining the output of 3–4 engineers as a single engineer through AI-assisted development — 1,000+ production changes landed over 2.5 years
- Eliminated ~52,000 redundant data operations per year, removing an estimated 2–4 FTE of recurring manual toil (~\$500K–\$1M annually in engineering time, storage, and metadata overhead)
- Achieved this leverage at under \$1,000 in total LLM spend — roughly 500x return on inference cost — by optimizing on cost per landed change (~\$1) rather than cost per API call
- Cut LLM input tokens ~60% via cacheable prompt prefixes and routed simple transforms to zero-cost local engines, holding per-change cost flat as volume scaled
- Cut manual cross-namespace data imports 90%+ (~157/day → ~14/day), driving the final phase of a multi-year warehouse consolidation from 80% to 100% coverage
- Unblocked the last ~20% of warehouse data — the segment no prior effort could touch because it sat in the direct ads revenue path, where a single day of degradation carries seven-figure impact — and delivered it with zero revenue loss
- Reduced engineer ramp-up from 2 weeks to 2 days and per-asset cycle time 3–5x by building a reusable AI agent playbook, with every hard-block gate derived from a production postmortem
- Operated under a constraint where errors surface as revenue loss rather than test failures, which drove the guard-first design: proved the pre-deployment validation stack was structurally blind — permission checks passed for tables that did not exist — and built the system that closed the gap
- Collapsed an entire failure class into an offline-checkable invariant: 81% of migrations required a namespace change, not the name change the tooling assumed
- Owned incident response on revenue-critical infrastructure — 5 production incidents including 2 high-severity — resolved fix-forward with zero reverts and permission grants coordinated across 6+ teams

Data Engineer

GREE International

2014 – 2015

PII masking, Vertica → Redshift, 1,000+ table optimization

Data Architect

Chegg Inc.

Oct 2013 – Apr 2014

Technical Director

Model N

2011 – 2013

Life Sciences BI, cloud migration, ETL modularization

Architect

CallidusCloud

2005 – 2011

Incentive comp analytics, BusinessObjects XI

DW Architect

Hewlett Packard

2001 – 2005

Enterprise DW + 8 datamarts; ETL 18 → 3 hrs; 40–50% sales lift via clickstream

Data Architect / Sr. Engineer

Xoriant · Dept of Electronics, India

1990 – 2001

Lead Data Architect — Partner Data Engineering

Jan 2022 – Jan 2024

VMware

Strategic architect for VMware's Partner Data Platform — unifying fragmented partner data into a single governed analytics layer

Data Architecture & Platform Unification

- Designed and built the Partner Data Platform from the ground up, unifying fragmented partner data from 10+ disparate sources into a single governed analytics layer — eliminating data silos that had blocked cross-functional decision-making for years
- Designed an intuitive data catalog and governance model (Python/Confluence) that drove broad self-service adoption, cutting reliance on tribal knowledge by 60% and reducing ad-hoc data request volume to the data team by ~40%
- Implemented VMware's first structured data governance for partner data — classification, lineage tracking, and policy enforcement — achieving audit-ready status and enabling the team to pass compliance reviews without remediation
- Improved data consistency by 60% through standardized naming conventions, automated quality checks, and cross-team data contracts — directly reducing downstream reporting errors that had caused quarterly restatements

AI & Performance Engineering

- Pioneered an AI copilot for retrieval optimization, coding, and governance best practices — one of VMware's earliest LLM-assisted data engineering deployments, reducing routine query development time by ~30%
- Achieved 40%+ Spark processing time improvements through broadcast joins, caching strategies, and skew management — enabling SLA-critical reports to complete within their delivery windows for the first time
- Established automated monitoring and proactive alerting that replaced reactive firefighting, reducing pipeline incident response time from hours to minutes and cutting weekend on-call escalations significantly

Leadership & Cross-Functional Alignment

- Drove cross-team alignment across 5+ data teams via shared roadmaps, reducing duplicate pipeline development efforts by ~25% and accelerating delivery cycles
- Led workshops and knowledge-sharing sessions that upskilled 30+ partner data consumers, building organizational data literacy and reducing the "data team as bottleneck" pattern
- Engaged director/VP-level stakeholders to define a unified data vision, securing budget and headcount for the platform roadmap — resulting in the team growing from 3 to 8 engineers during the engagement

Data Engineer (Consultant)

Oct 2015 – Dec 2021

Meta

Performance, query-engine migration and reliability across Meta's ads, integrity, commerce and marketing data

Performance & Cost Engineering

- Cut a critical ads pipeline from 5 hours to under 1 hour by decomposing serial split tasks into 30 parallel threads — recovering ~4 hours of daily batch window on a revenue-reporting dependency
- Brought a pipeline blocked at 10× its CPU budget back within quota through query optimization, unblocking a stalled finance reporting flow
- Cut data scanned per run ~30× on a high-cardinality user dimension check by narrowing a full-month partition scan to a single day
- Resolved repeated evictions from CPU overage and eliminated recurring out-of-memory failures on high-volume ad conversion fact tables through timestamp-partition parallelization with signal-based synchronization

Query Engine Migration & Judgment

- Migrated 89 production pipelines from Hive to Spark and Presto over 11 months (~8/month sustained) — datelist, cube and funnel workloads across ads, integrity, commerce and marketing
- Benchmarked rather than assumed: on one workload Spark failed 40 times and ran 50+ minutes while Presto finished in under 30 — shipped Presto with documented evidence, avoiding a slower, failure-prone pipeline
- Extended the funnel framework to support Spark as a query engine, making the migration path reusable for other teams rather than a one-off conversion
- Tuned Spark execution profiles — dynamic allocation, bucketing, partition-strategy overrides, executor and driver memory — resolving conversions that failed on JVM heap limits rather than query logic

Commerce, Privacy & Reliability

- Designed and launched the Category Management Data Warehouse and owned the SMB funnel redesign, improving query performance 3x while scaling to tens of thousands of small-business advertisers without infrastructure changes
- Led privacy remediation across 1,200+ SMB 2.0, Customer Journey and BPO tables, and drove Salesforce ID deprecation across 1,000+ pipelines — eliminating an external vendor dependency affecting ~15% of cross-system joins
- Carried pipeline oncall across ~200 incidents with root-cause fixes rather than restarts; delivered ~50 privacy and compliance remediations and reduced storage cost across ~60 retention and cleanup engagements
- Built chargeback representment and leakage dashboards, and a Marketplace reliability dashboard covering payins/payouts — surfacing previously undetected revenue leakage and cutting MTTR by replacing manual log analysis
- Recognized by 45 colleagues across 53 peer commendations for debugging, escalation and cross-team unblocking across ads, payments, commerce, marketplace and privacy

ADDITIONAL RECENT EXPERIENCE

Data Engineer Consultant

Dec 2017 – Apr 2018

LinkedIn

- Led GDPR compliance efforts: encrypted sensitive customer data into LinkedIn Dali storage using Hive, Python, and Pig; retired legacy data sources and feeds; migrated to a stable, compliant framework